



SE 102.3

Web Based Application Development

Project Report

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Group J

23.2 FOC

<http://warehouse.000.pe>

Abstract

This project report presents the development of an online shopping website specifically designed for an electronic items shop. The website provides a seamless and user-friendly platform where customers can browse, select, and purchase a wide range of electronic products. Built using HTML, CSS, JavaScript, and PHP, the website integrates essential e-commerce functionalities such as user account creation, wish list management, shopping cart, and secure online payment processing.

Additionally, the platform includes advanced features like order tracking, customer reviews to enhance user experience and satisfaction. The project aims to deliver a reliable and efficient online shopping solution, catering to the growing demand for digital transactions in the electronics market.

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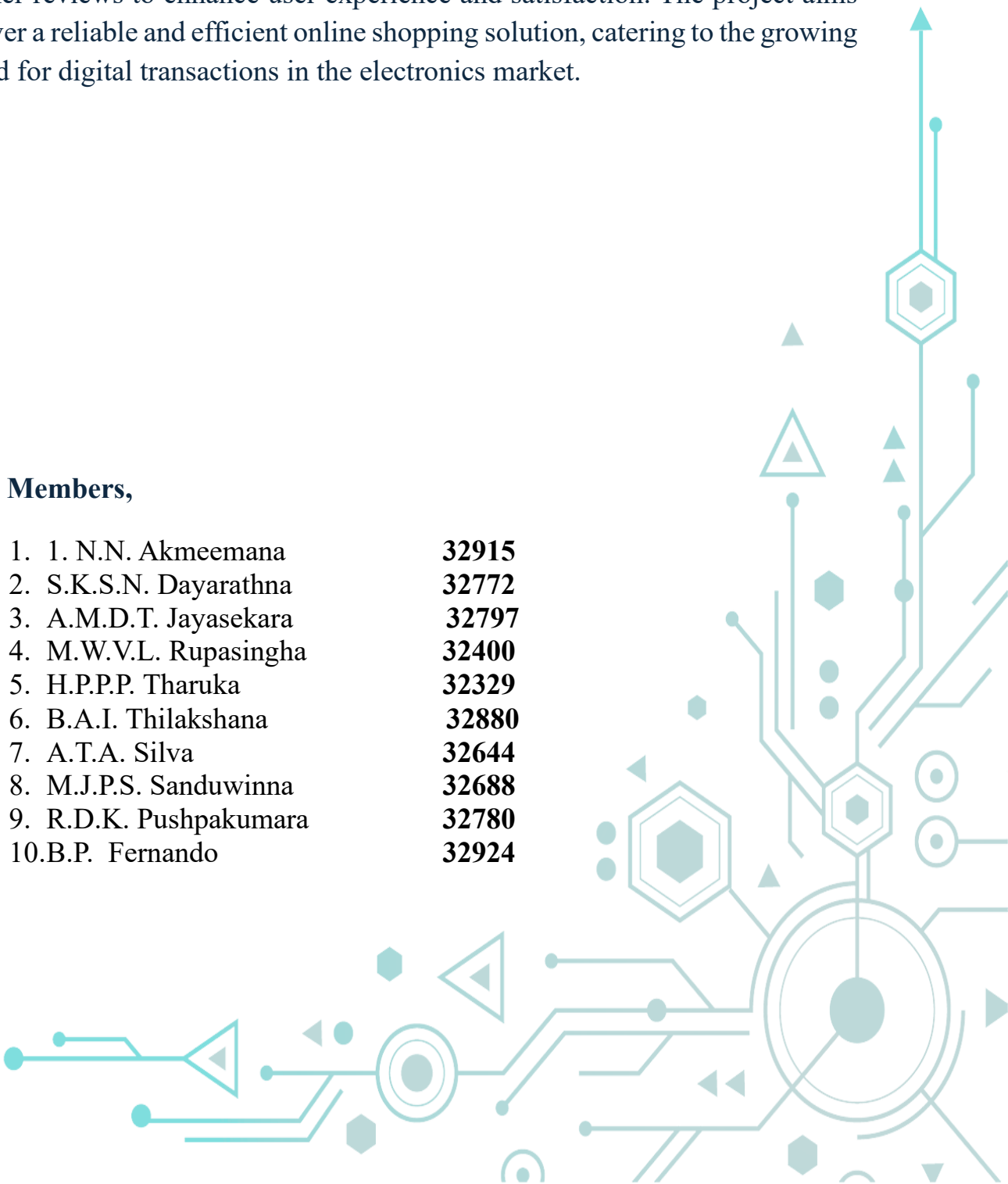


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Introduction

Online shopping has drastically changed the retail industry in the modern digital era by providing customers unparalleled ease and an extensive selection of goods. Because of the growing demand for online platforms, especially in the electronics industry, companies are developing advanced e-commerce websites. The development of an online store specializing in electronics is the main goal of this project, which aims to offer customers and sellers in the electronic goods industry a complete solution.

This project's main goal is to create and deploy a simple online shopping environment where users can easily browse, select, and buy electronic products. The capability for users to make accounts, maintain wish lists, add products to shopping carts, secure online payment methods and more are important features.

The website has a strong backend for handling data and transactions and a responsive, interactive user experience thanks to the usage of HTML, CSS, JavaScript, and PHP in its construction. Developing essential features like user authentication, product catalog administration, and payment gateway integration are all included in the scope. Within the project's scope are considerations for future scalability and the potential for additional services, such as AI-based 24/7 customer assistance and customized recommendations.

Technology Stack

The development of the online shopping website for an electronic items shop leverages a combination of front-end and back-end technologies to create a responsive, interactive, and secure platform. The choice of these technologies ensures that the website not only delivers a smooth user experience but also maintains robust performance and security.

Front-End Technologies

-HTML (HyperText Markup Language): HTML forms the backbone of the website, providing the basic structure and layout of the web pages. It is used to create and organize content, text, images, and forms, that are displayed in the user's browser.

-CSS (Cascading Style Sheets): CSS is used to style the HTML elements. It handles the layout, colors, fonts, and overall aesthetic of the website, ensuring a consistent and user-friendly interface across different devices and screen sizes.

-JavaScript: JavaScript is used to add interactivity and dynamic content to the website.

Back-End Technologies

-PHP (Hypertext Preprocessor): PHP is server-side scripting language responsible for handling the logic behind the website's functionality. It interacts with the database to manage user accounts, process transactions, store order details, and retrieve product information.

-MySQL: A relational database management system used for storing and managing data related to users, products, orders, and reviews. It ensures efficient data retrieval and manipulation.

Version Control

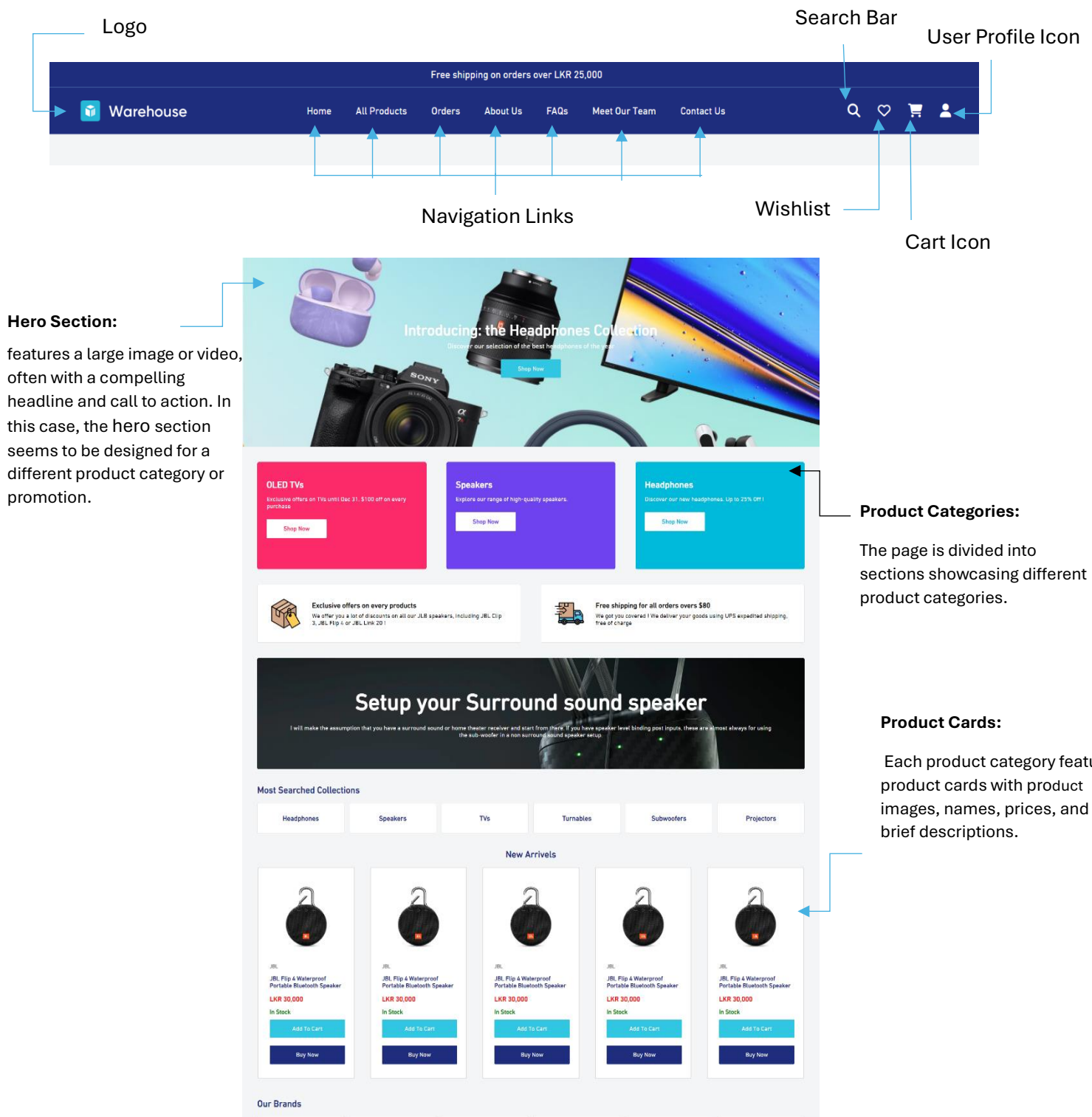
-Git: A version control system used to track changes in the codebase and facilitate collaboration among team members.

-GitHub: A web-based platform used for hosting the project's repository, enabling version control and collaborative development.

System Design

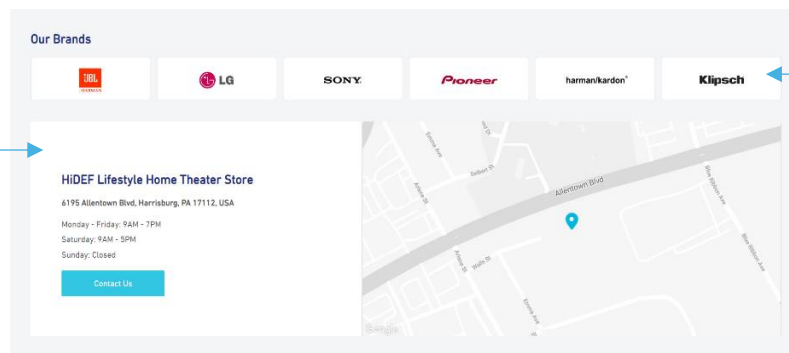
User Interface Design:

The website has a simple, modern appear, and its layout and design work effectively to promote and sell electronic goods. A good user experience is produced by focusing product presentation, providing simple navigation, and having a visually appealing layout. Because of the layout's organization, users may easily navigate and find whatever data they require.



Store Information and Location:

A Google Maps integration provides a visual representation of the store's location, making it easy for customers to find the store.



Brand Logos:

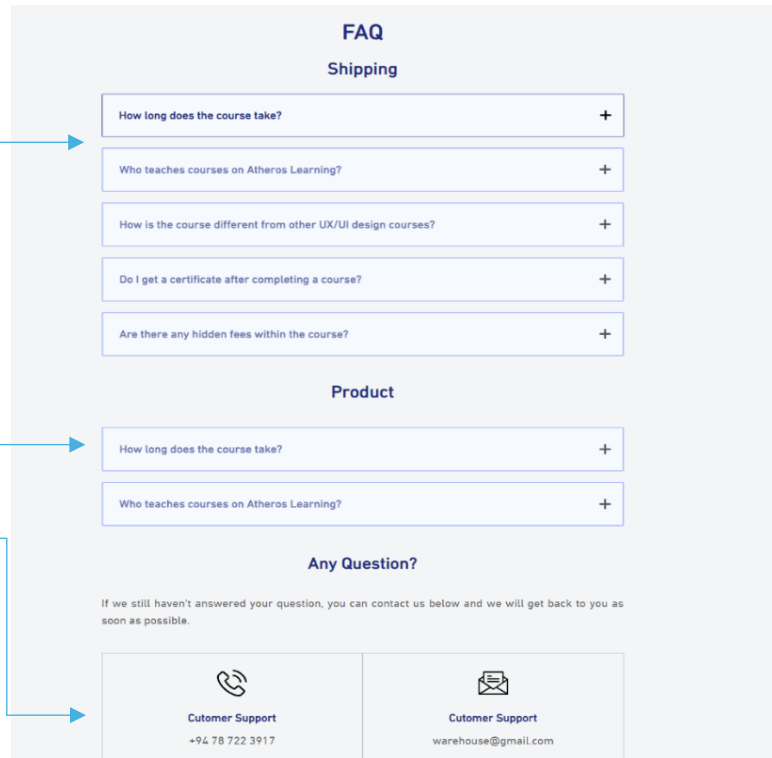
Displayed prominently, immediately conveying the range of brands available.

Accordion-Style Questions:

This structure helps users quickly find the information they need.

Contact Information:

The store's contact information is provided, making it easy for customers to reach out for further assistance.

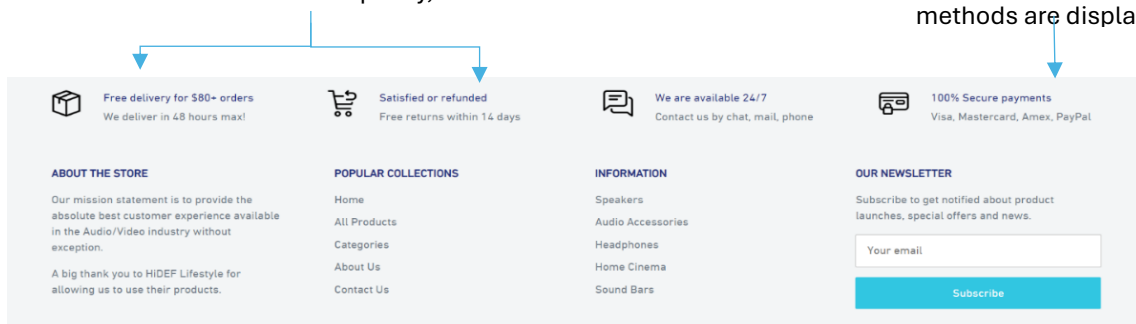


Customer Guarantees:

The section highlights the store's commitment to customer satisfaction with guarantees like free delivery, satisfied or refunded policy, and free returns

Payment Options:

The accepted payment methods are displayed



Newsletter Signup:

The section includes a newsletter signup form, allowing customers to stay updated on product launches, offers, and news.

Database Structure:



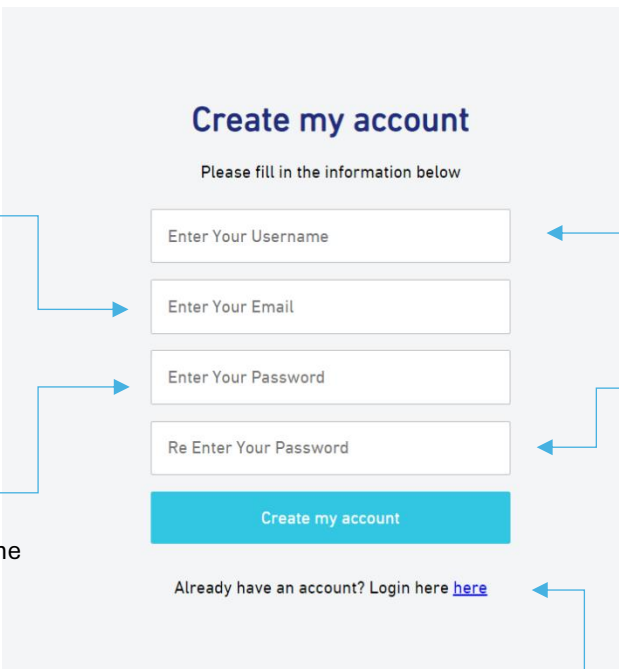
Table	Action	Rows	Type	Collation	Size	Overhead
admins	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	16.0 KiB	-
cart	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	16.0 KiB	-
messages	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	16.0 KiB	-
orders	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	16.0 KiB	-
products	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	16.0 KiB	-
users	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	16.0 KiB	-
wishlist	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
7 tables	Sum	19	InnoDB	utf8mb4_general_ci	112.0 KiB	0 B

The website's robust backend, programmed in PHP. Database, named “shop_db,” is designed to efficiently store and retrieve data. The database consists of several tables, each serving a specific purpose:

- **Admins Table:** Stores administrator credentials and profile information.
- **Cart Table:** Holds items currently selected by users for purchase.
- **Messages Table:** Contains communication between customers and support.
- **Orders Table:** Logs details about customer purchases, including order status and payment information.
- **Products Table:** Contains all information about items for sale, such as product name, description, price, and stock levels.
- **Users Table:** Keeps track of registered users' details, including their login credentials and contact information.
- **Wishlist Table:** Allows customers to save desired products for future consideration.

Features and Functionality

User Registration and Login:



Create my account

Please fill in the information below

Email Address:
A valid email address for communication and account verification

Username:
A unique identifier for the user's account.

Password:
A strong password to protect the user's account.

Password Confirmation:
Re-entering the password to ensure accuracy.

Enter Your Username

Enter Your Email

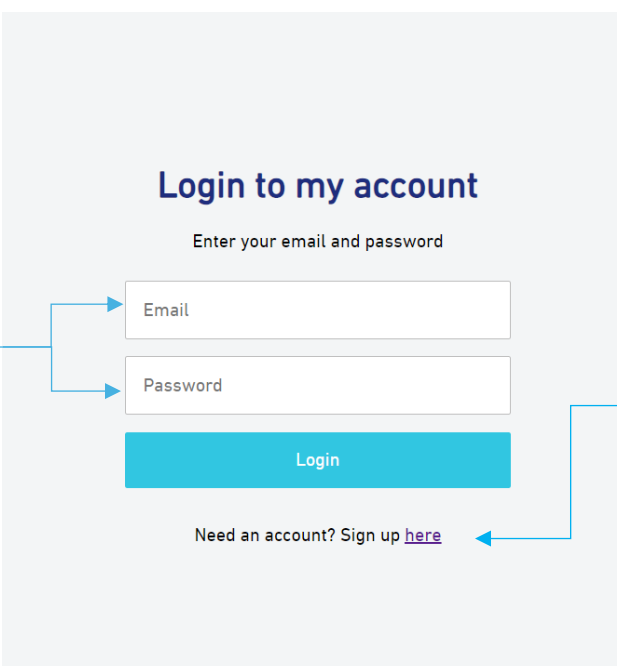
Enter Your Password

Re Enter Your Password

Create my account

Already have an account? Login here [here](#)

This link directing users to the login page.



Login to my account

Enter your email and password

User can easily login their account using email and password.

*User passwords are encrypted before being stored in the database.

Email

Password

Login

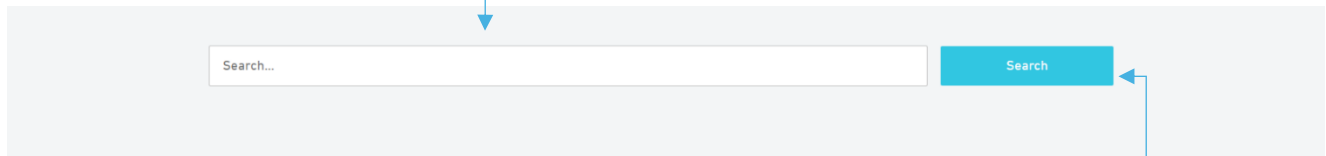
Need an account? Sign up [here](#)

A link is provided for users who do not have an account to create one.

*Users are notified of incorrect login details and provided with options to recover their accounts.

Product Catalog:

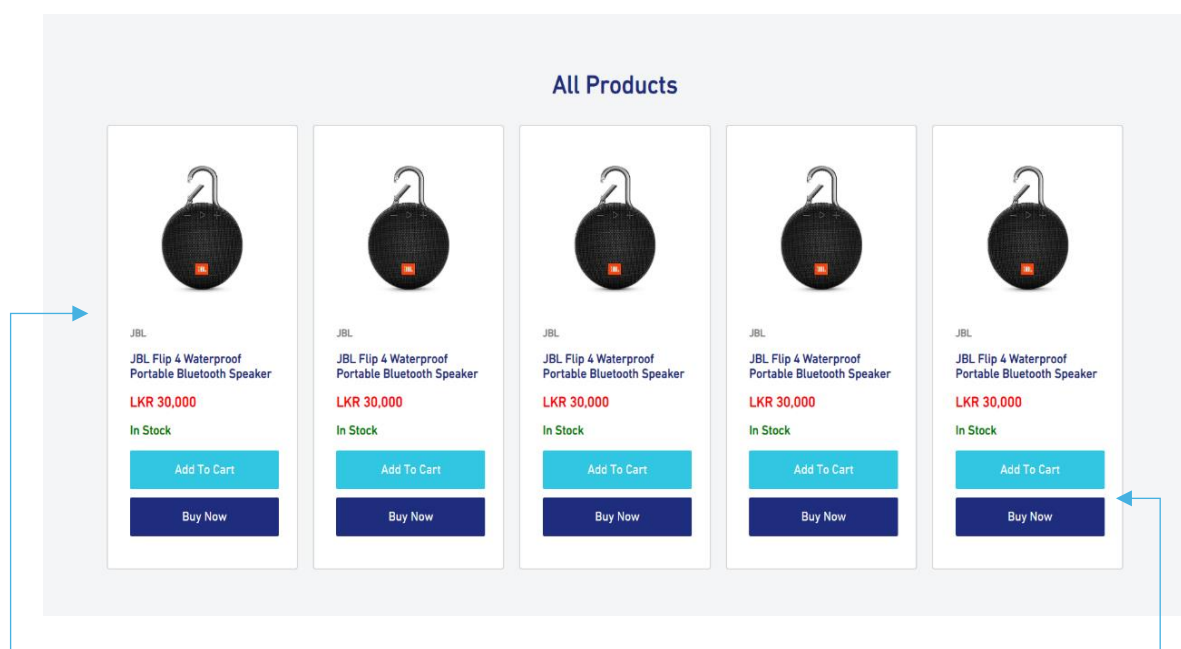
Users can product names into the search bar to find relevant products.



When user clicks the "Search" button or presses the Enter key to initiate the search.

Then website displays a list of products that match the search query.

******ensures that customers have all the necessary information at their fingertips, enhancing their shopping experience.

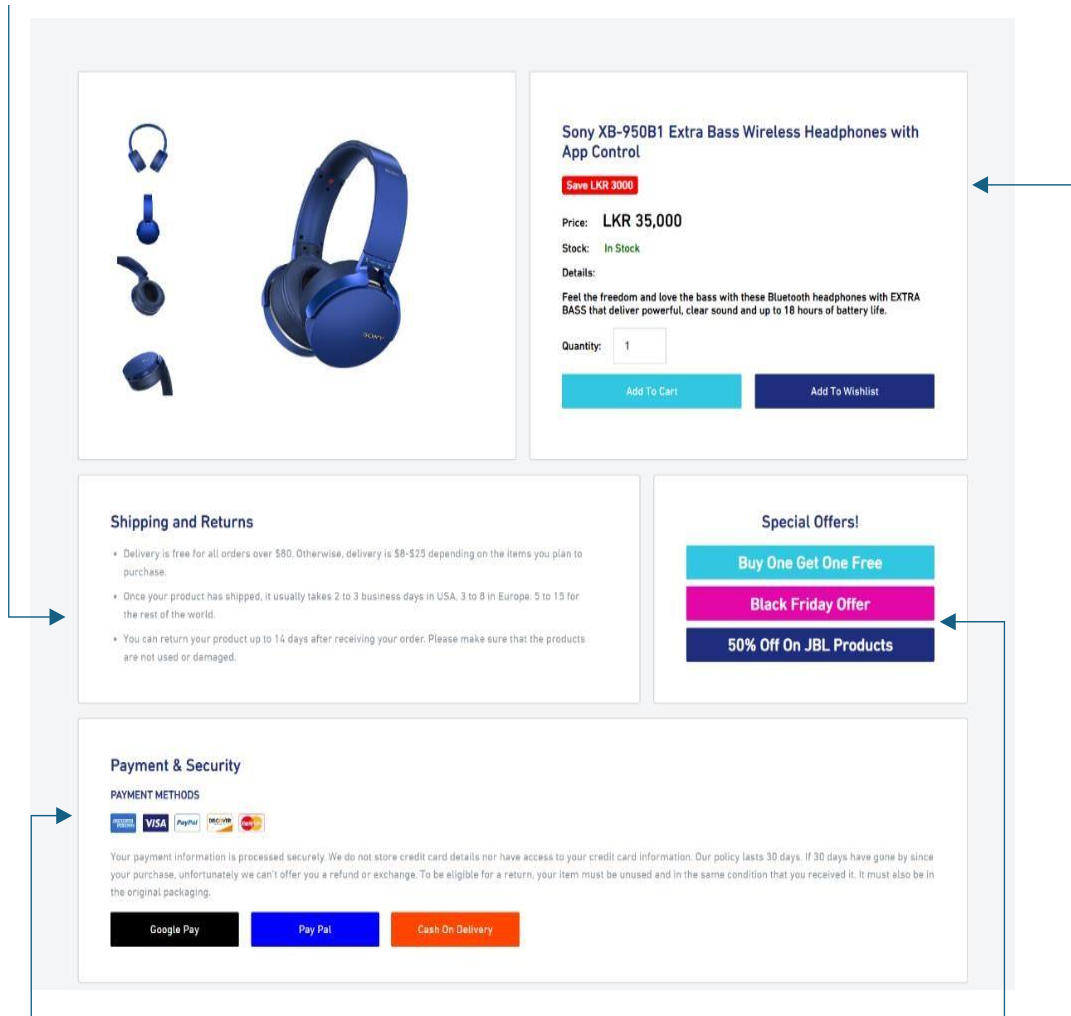


- A clear and visually appealing image of the product.
- A concise and informative product name.
- The product's price often displayed prominently.

"Add to Cart" and "Buy Now" that allow users to take immediate action.

Information about shipping times and return policy.

Each item with detailed information to help customers make informed purchasing decisions.

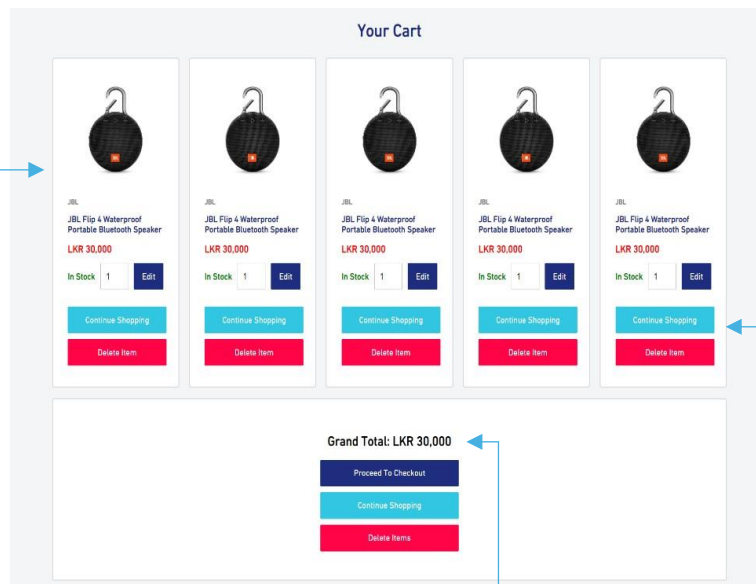


Ensures customers that their payment information is secure.

Current promotions and discounts.

Shopping Cart and Checkout:

******User can view their cart by clicking on the cart icon or through direct navigation links.



Displays all selected items, their quantities, and individual prices

Options to edit quantities and remove items are

The cart automatically calculates the total cost, including item prices, taxes, and any applicable shipping fees based on the customer's location and shipping preferences.

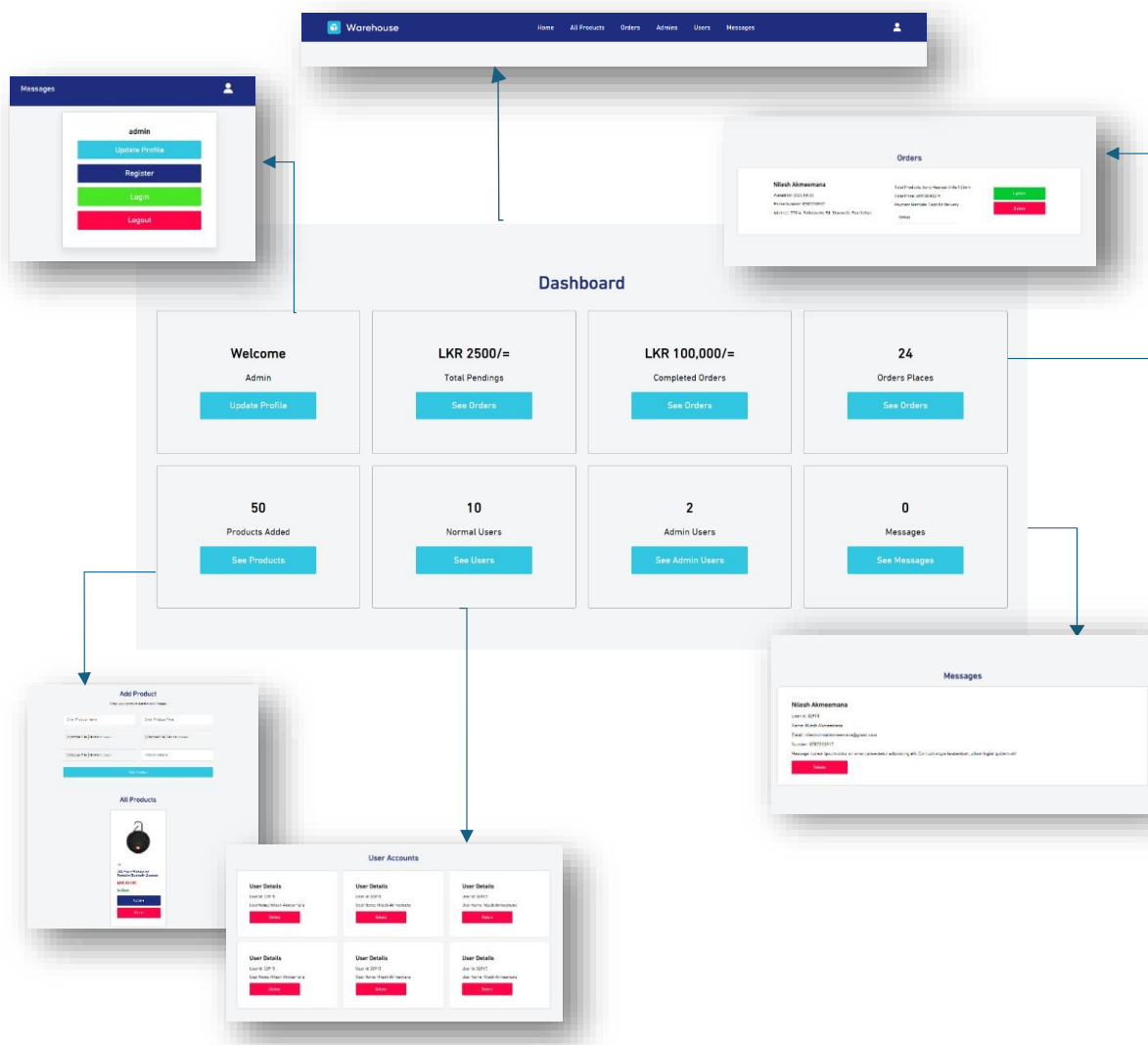
The checkout process involves entering delivery details, selecting a shipping method, and providing payment information.

The screenshot shows a 'Checkout' page with the heading 'Place Your Order'. It contains several form fields arranged in two columns. The left column includes fields for 'Enter Your Name', 'Enter Your Email', 'Address Line 01', 'Enter Your City', 'Enter Your Country', and 'Enter Your Zipcode'. The right column includes fields for 'Enter Your Number', 'Payment Method', 'Address Line 02', and 'Enter Your State'. At the bottom of the left column is a large blue 'Place Order' button. To the right of the form fields is a box titled 'Your Orders' which displays 'Grand Total: LKR 30,000' and a list of five items: 'Sony Headset 300ed (52500/- x 1)'. A blue arrow points from the explanatory text block to the 'Place Order' button, and another blue arrow points from the same text block to the 'Your Orders' summary box.

A summary of the order is displayed for review before finalizing the purchase.

If order is confirmed, customers receive an order confirmation on the website and via email, detailing their purchase and expected delivery times.

Admin Panel:



- Admin Profile:** Personalize admin account settings, update contact information or password.
- Managing Products:** Add new items, edit existing product details, and remove listings as needed.
- Managing Users:** Oversee customer accounts, approve registrations, and handle user queries or complaints.
- Managing Orders:** Track order status from pending to completed, total number of orders placed and manage order details.
- Messages:** View and respond to customer messages and inquiries.

******These features ensure that administrators can efficiently manage the website, maintain product listings, oversee user activities, and handle orders effectively.

Implementation

For to ensure a flawless user experience and effective platform management, modern tools and technologies were strategically included during the website's development and implementation. This is a detailed rundown of all the tools and processes used in the implementation stage.

- Visual Studio Code (VSCode) IDE and Collaboration As the main Integrated Development Environment (IDE), VSCode improved the coding experience with its user-friendly interface and broad support for plugins.
-GitHub The utilization of GitHub for collaborative development was crucial since it allowed team members to easily synchronize code changes.
- Git source control Git was used as the source control platform, which made version tracking and team work easier. Code management was improved and effective teamwork was guaranteed through integration with VSCode.
- XAMPP, the local development environment XAMPP was used as the local development environment; it served PHP files locally and used MariaDB to host the database. This made it easier to emulate a server environment, which is essential for thorough testing and problem solving.

Launching the Website

Hosting and File Transfer Management:

The website for WAREHOUSE was hosted by InfinityFree, a free web hosting service renowned for its reliable service and zero-cost deployment, in order to make it accessible to everyone. InfinityFree is a great option for starting WAREHOUSE because it has the power to host small to medium-sized websites. You can access the website at <http://warehouse.000.pe>.

Collaboration and Version Control:

GitHub, a popular version control tool, was utilized to manage and collaborate on the project during the development stage. Team members were able to monitor changes, add to the coding, and efficiently oversee the project because of the GitHub project repository WAREHOUSE. You can find the repository at GitHub Repository.

<https://github.com/nileshakmeemana/Web-Project-Group-J-.git>

Challenges and Solutions

Several challenges had to be faced all over the development of the WAREHOUSE website, but each one was handled carefully and thoughtfully in order to develop a reliable, safe, and easy-to-use online service platform.

protecting personal data by guaranteeing website security. For securing sensitive data, we used secure coding techniques, password hashing, and SSL encryption.

Our team used a structured and modular coding strategy to ensure code maintainability. Git and other version control technology allowed for collaborative development while maintaining code integrity and simplifying management. proving our dedication to improving overall maintainability and code structure.

The design and optimization of databases created specific issues that called for specialist solutions. Efficient data retrieval was achieved by creating a normalized database schema, optimizing database queries, and appropriately indexing columns. To maintain top performance over time, regular optimization and analysis methods were implemented.

Modern JavaScript features, polypills, and feature detection were all included in the strategy to address browser compatibility with JavaScript.

Future Enhancement

The online shopping website might benefit from a number of changes to increase functionality and user experience as technology develops and customer expectations rise.

- **Advanced Analytics Dashboard** - Make personalized product recommendations based on user behavior, past purchases, and preferences by implementing artificial intelligence algorithms.
- **Mobile Application Development** - To provide users a flawless shopping experience on their smartphones and tablets, develop a mobile application for the Android and iOS platforms.
- **Enhanced Security Features** - While the current website has some basic security measures in place, further work might concentrate on including advanced security features, like data encryption, two-factor authentication, and frequent security audits to protect user information and transactions.
- **Localization and Multilingual Support** - By including these capabilities, the website will be able to serve a worldwide user base, support a wider variety of languages, and enhance accessibility for people who do not understand English.
- **Expanded Payment Options** - Offering more payment options, such as cryptocurrencies, digital wallets, and buy-now-pay-later services, would appeal to a wider customer base and enhance the checkout process.

It is possible for the e-commerce website to stay competitive in the e-commerce business and meet the dynamic needs of the market by following these future developments.

Conclusion

In summary, with the future process of creating the WAREHOUSE website serves as a tribute to our combined successes, knowledge gained, and the priceless contribution of every team member. The project's success is evidenced by the seamless integration of multiple functions, like user authentication on major websites, dynamic order systems, and adaptable dashboards for administrators and users.

During the development process, we overcame several challenges and improved our methods for handling security concerns, responsive design, and cross-browser compatibility. These difficulties increased awareness of the complexity of web development and offered opportunities for future growth.

Advanced tools like GitHub and Visual Studio Code were integrated to speed up deployment times and enable collaborative development. The project taught us several valuable lessons, including the value of thorough planning, excellent teamwork, and the necessity of constant challenge adaptation. Accepting a modular code structure and giving user experience principles top priority were key realizations that greatly influenced the WAREHOUSE website's success.

Finally, we would like to express our sincere gratitude to every team member who contributed their knowledge, skills, and imagination to making WAREHOUSE possible and especially thank the university teachers and mentors for their guidance and support during the development process. Looking back on this adventure, we have a plethora of knowledge and abilities that will surely influence our web application development activities in the future.

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